Rule identifiability in elementary cellular automata under dynamical noise

Frank Peterlein*

Version 0.2, November 2026 — extends version 0.1 of 15 October 2026

Abstract

How well can the rule of an elementary cellular automaton be read off a trajectory when the dynamics itself is noisy? On a ring of $n = 12$ cells the noisy automaton is a Markov chain on 4096 states, so every quantity in Sections 1–4 is computed exactly: no sampling, no seeds; Section 5 extends the exact small-noise analysis to $n = 4, \dots, 20$. Without noise, identifiability is combinatorial. A trajectory fixes the rule on the neighbourhoods it visits and on nothing else; the visited set stops growing once the orbit closes; over all 256 rules and 4096 initial states, 66.9% of the pairs determine the rule and the remaining pairs never do. With dynamical noise every trajectory law has full support and identifiability becomes a rate: the evidence against the hardest alternative grows linearly in time, at $\rho(f;\varepsilon) = \lambda(\varepsilon)\,\min_c \nu_c(f;\varepsilon)$ nats per step, the product of a pure noise factor and the stationary frequency of the rarest neighbourhood. For 42 of the 88 rule classes some neighbourhood occurs on no cycle of the deterministic map, so ρ vanishes at both ends of $[0, \frac{1}{2}]$ and the best noise level is interior; 18 further classes behave the same way; class 106 falls over the computed grid, but its rate diverges as $\varepsilon \to 0$ (Section 5); 27 keep rising as $\varepsilon \to 0$. For the 61 classes that fall over the grid we locate the best noise level, $\varepsilon^* \in [0.010, 0.293]$ (for class 106, which attains the lower end, a local maximum only). Noise therefore does not help uniformly: it helps exactly those rules whose noise-selected attractors exercise only part of the rule table. At finite horizon the same mechanism rescues initial states that are permanently non-identifiable without noise. A boundary check with generators outside the family shows the flip side: the observer converges confidently to the nearest rule in its family, while the misfit becomes visible at a lower rate. Version 0.2 adds the exact small-noise theory: the order of every stationary probability is an integer computed from the stochastic potentials of the cycles, and $\rho(f;\varepsilon) \sim a\,\varepsilon^{k^*} \log \frac{1}{\varepsilon}$ with an exactly computed order k^*. At $n = 12$ this settles the 46 classes left open by version 0.1: 28 have $k^* = 0$ ($\rho \to \infty$) and 18 have $k^* \geq 1$; one class (106) changes side. For $n = 4, \dots, 20$ the classification is computed for all classes; for $n \geq 8$, 13 classes change side with the ring size. The noise-selected cycles at $n = 17$ are already determined by the single-flip attractor chain of Bagnoli et al. (2024); this version adds their exact orders.

1 Definitions

1.1 Setting. Fix $n \geq 3$ (computed case: $n = 12$). States are $x \in S = \{0,1\}^n$ on a ring. An elementary rule is a map $f : \{0,1\}^3 \to \{0,1\}$; $\mathcal{R}$ is the set of all 256 (Wolfram, 1983). The global map is $(F_f x)_i = f(x_{i-1}, x_i, x_{i+1})$, indices mod n. The neighbourhood set of a state is $N(x) = \{(x_{i-1}, x_i, x_{i+1}) : i\} \subseteq \{0,1\}^3$; we write $c \in \{0, \dots, 7\}$ for the code $4x_{i-1} + 2x_i + x_{i+1}$ and $n_c(x)$ for the number of positions of x with code c, so that $\sum_c n_c(x) = n$.

1.2 Dynamics. For $\varepsilon \in [0, \frac{1}{2}]$: $X_0 = x_0$, $X_{t+1} = F_f(X_t) \oplus Z_t$, with Z_t i.i.d. vectors of independent Bernoulli(ε) bits. The transition kernel is

$$P_{f,\varepsilon}(y \mid x) = \varepsilon^{H(y, F_f x)} (1-\varepsilon)^{n - H(y, F_f x)},$$

*Independent researcher, Berlin. ORCID 0009-0002-1886-1303. Contact: `frnkptrln` on GitHub. See the statement on the use of AI at the end of the note.

H the Hamming distance. Noise acts on the state and propagates; it is not a noisy readout of a deterministic orbit (Remark 1).

1.3 Observation and hypotheses. The observer sees the full trajectory $x_{0:T}$ and knows $\mathcal{R}$, ε and x_0. The unknown is f. The object of study is the family of trace laws $P^T_{f,\varepsilon}(\cdot \mid x_0)$ on S^T, $f \in \mathcal{R}$.

1.4 $\varepsilon = 0$: combinatorial identifiability. Let x_t be the orbit of f from x_0 and $N_T(f, x_0) = \bigcup_{t<T} N(x_t)$. The consistent set is

$$C_T(f, x_0) = \{g \in \mathcal{R} : F_g(x_t) = x_{t+1}\ \forall t < T\} = \{g : g = f \text{ on } N_T(f, x_0)\}, \qquad |C_T| = 2^{\,8-|N_T|}.$$

For fixed (x_0, T) this is an equivalence relation on $\mathcal{R}$. Since the orbit is eventually periodic, N_T is non-decreasing in T and constant for $T \geq \tau(f, x_0)$ (transient plus period). The class therefore never shrinks below $2^{8-|N_\infty|}$: after τ, watching longer buys nothing.

1.5 $\varepsilon > 0$: quantitative identifiability. For $0 < \varepsilon < \frac{1}{2}$ every trace law has full support on S^T, so $C_T = \mathcal{R}$ and identifiability must be graded. Define the *evidence* against g when f is true,

$$E_T(f\|g; x_0, \varepsilon) = \mathrm{KL}\big(P^T_{f,\varepsilon}(\cdot \mid x_0) \,\big\|\, P^T_{g,\varepsilon}(\cdot \mid x_0)\big) = \mathbb{E}_f\left[\log \frac{P_f(X_{1:T} \mid x_0)}{P_g(X_{1:T} \mid x_0)}\right].$$

Lemma 1 (decomposition)**.** *With $d_{fg}(x) = H(F_f x, F_g x)$ and $\lambda(\varepsilon) = (1-2\varepsilon)\log\frac{1-\varepsilon}{\varepsilon}$,*

$$E_T(f\|g; x_0, \varepsilon) = \lambda(\varepsilon) \sum_{t=0}^{T-1} \mathbb{E}_f[\, d_{fg}(X_t)\,].$$

Proof. Chain rule of relative entropy along the Markov chain (Cover and Thomas, 2006, Thm. 2.5.3): $E_T = \sum_t \mathbb{E}_f[\mathrm{KL}(P_f(\cdot \mid X_t)\|P_g(\cdot \mid X_t))]$. Given x, $\log \frac{P_f(y|x)}{P_g(y|x)} = \big(H(y, F_g x) - H(y, F_f x)\big)\log\frac{1-\varepsilon}{\varepsilon}$; bits where $F_f x$ and $F_g x$ agree contribute 0, each of the $d_{fg}(x)$ bits where they differ contributes $+1$ with probability $1-\varepsilon$ and -1 with probability ε, i.e. $1-2\varepsilon$ in expectation. □

Margin and rate. $m_T(f; x_0, \varepsilon) = \min_{g \neq f} E_T(f\|g; x_0, \varepsilon)$ is the evidence against the hardest alternative. For $\varepsilon > 0$ the kernel is strictly positive, hence the chain is ergodic with a unique stationary law $\pi_{f,\varepsilon}$ of full support (Norris, 1997), and $E_T/T \to \lambda(\varepsilon)\mathbb{E}_{\pi_{f,\varepsilon}}[d_{fg}]$. The *identifiability rate* of a rule is

$$\rho(f; \varepsilon) = \min_{g \neq f} \lambda(\varepsilon)\, \mathbb{E}_{\pi_{f,\varepsilon}}[\, d_{fg}\,],$$

independent of x_0. ρ is constant on the 88 classes of rules under reflection and complementation (Li and Packard, 1990), since the chain is equivariant under both.

Lemma 2 (the hardest alternative is a one-bit neighbour)**.** *Let $\nu_c(f; \varepsilon) = \mathbb{E}_{\pi_{f,\varepsilon}}[n_c]$ be the expected number of positions with code c under the stationary law. Then $\mathbb{E}_\pi[d_{fg}] = \sum_c \nu_c\, \mathbf{1}[f(c) \neq g(c)]$, and*

$$\rho(f; \varepsilon) = \lambda(\varepsilon)\, \min_c \nu_c(f; \varepsilon),$$

the minimum being attained by the rule that differs from f only on the rarest code. The same holds termwise at finite horizon: $m_T(f; x_0, \varepsilon) = \lambda(\varepsilon) \min_c \sum_{t<T} \mathbb{E}_f[n_c(X_t)]$.

Proof. $d_{fg}(x)$ counts the positions whose code lies in the set where f and g differ, so $d_{fg}(x) = \sum_c n_c(x)\mathbf{1}[f(c) \neq g(c)]$; take expectations. The sum over a non-empty set of non-negative terms is at least its smallest term, with equality for a singleton. □

Lemma 3 (ends of the curve)**.** $\rho(f;\frac{1}{2}) = 0$ *because* $\lambda(\frac{1}{2}) = 0$*. As* $\varepsilon \to 0$*, every limit point of* $\pi_{f,\varepsilon}$ *is a stationary law of the deterministic map, i.e. a mixture of the uniform laws on the cycles of* F_f*. If some code* c *occurs in no state of any cycle of* F_f*, then* $\nu_c(f;\varepsilon) = O(\varepsilon)$ *while* $\lambda(\varepsilon) = O(\log \frac{1}{\varepsilon})$*, hence* $\rho(f;\varepsilon) \to 0$*: the curve vanishes at both ends and attains its maximum in the interior of* $(0, \frac{1}{2})$*; we write* $\varepsilon^*(f)$ *for a maximiser (uniqueness is not asserted). If instead every code occurs on some cycle that carries positive weight in the small-noise limit,* $\min_c \nu_c$ *stays bounded away from* 0 *and* $\rho(f;\varepsilon) \to \infty$*.*

Proof. For the first statement: $\varepsilon \mapsto P_{f,\varepsilon}$ is continuous and the simplex is compact, so every limit point π_0 of $\pi_{f,\varepsilon}$ satisfies $\pi_0 = \pi_0 P_{f,0}$, and the stationary laws of a map are exactly the mixtures of uniform laws on its cycles. For the $O(\varepsilon)$ bound let A be the set of cycle states and $B = S \setminus A$, and let L be the longest transient. From any state the chain follows the deterministic map for one step with probability $(1-\varepsilon)^n \geq 1 - n\varepsilon$, so a sojourn in B ends within L steps unless a noise event occurs during those steps; the expected sojourn length is therefore at most $L/(1-n\varepsilon)^L$. A sojourn in B can only start by a noise event from A, which has probability at most $n\varepsilon$ per step. By the renewal structure of the stationary chain, $\pi_{f,\varepsilon}(B) \leq n\varepsilon\, L/(1-n\varepsilon)^L = O(\varepsilon)$. A code that occurs on no cycle occurs only in B, so $\nu_c \leq n\, \pi_{f,\varepsilon}(B) = O(\varepsilon)$. The last statement follows from $\nu_c \to \sum_{\text{cycles } C} w_C\, \overline{n_c}(C)$ with the limiting weights w_C (Freidlin and Wentzell, 2012). □

Which cycles carry weight in the small-noise limit is a question about the large-deviation costs between basins (Freidlin and Wentzell, 2012); version 0.1 did not compute those weights, Section 5 of this version does. It uses the checkable sufficient condition of Lemma 3 (a code absent from every cycle), and reports the numerical trend of ρ at the smallest resolved noise levels for the remaining classes.

Remark 1 (why the noise must be dynamical). If instead a readout channel K corrupts the deterministic orbit, the trace law is $K \circ P_{f,0}$, and by the data-processing inequality the evidence E_T can only decrease; in particular rules that the orbit does not separate ($g \in C_T(f, x_0)$, equal trace laws) remain inseparable under every readout channel. "Noise increases identifiability" is a statement about noise in the dynamics only.

Remark 2 (what the evidence licenses). For any decision rule, i.e. any event A, $P_f(A^c) + P_g(A) \geq \frac{1}{2}\exp(-E_T(f\|g))$ (Bretagnolle and Huber, 1979; Tsybakov, 2009). A small margin therefore bounds from below the confusion of f with its hardest alternative under *every* procedure; this is the direction that establishes non-identifiability. Upper bounds on the error of the maximum-likelihood decoder can be added via Chernoff exponents; this note reports E, m and ρ exactly and does not report decoder error rates.

1.6 Declared boundaries. The true rule lies in $\mathcal{R}$; ε is known to the observer; observation is complete; $n = 12$ on a ring (Section 5 also $n = 4, \ldots, 20$). Each is a boundary of the claim, not an experiment of this note; Section 4 probes the first of them with a check, not with a theorem.

2 Computation

Every number in Sections 1–4 is a function of the 4096-state chain and is computed as such; Section 5 computes exact small-noise orders for $n = 4, \ldots, 20$.

Stationary laws. For each rule and each ε on the grid $\varepsilon \in \{0.001, 0.002, 0.005, 0.01, 0.02, 0.03, 0.05, 0.075, 0.1, 0.125, 0.15, 0.2, 0.25, 0.3, 0.35, 0.4, 0.45, 0.5\}$ (18 values; at $\varepsilon = \frac{1}{2}$ the stationary law is uniform and $\rho = 0$ analytically), $\pi_{f,\varepsilon}$ is obtained by solving the linear system $\pi(I - P_{f,\varepsilon}) = 0$, $\sum \pi = 1$, with the dense 4096×4096 transition matrix. The solution is then verified by power iteration from three different initial laws (uniform, and the point masses at the all-zero and the all-one state): an entry is *resolved* when all three converge to the dense solution to within 10^{-6} in every ν_c; otherwise the chain is metastable at that ε (it mixes too slowly for the stationary law to be resolved numerically), and the entry is excluded from everything that follows. Over the 88

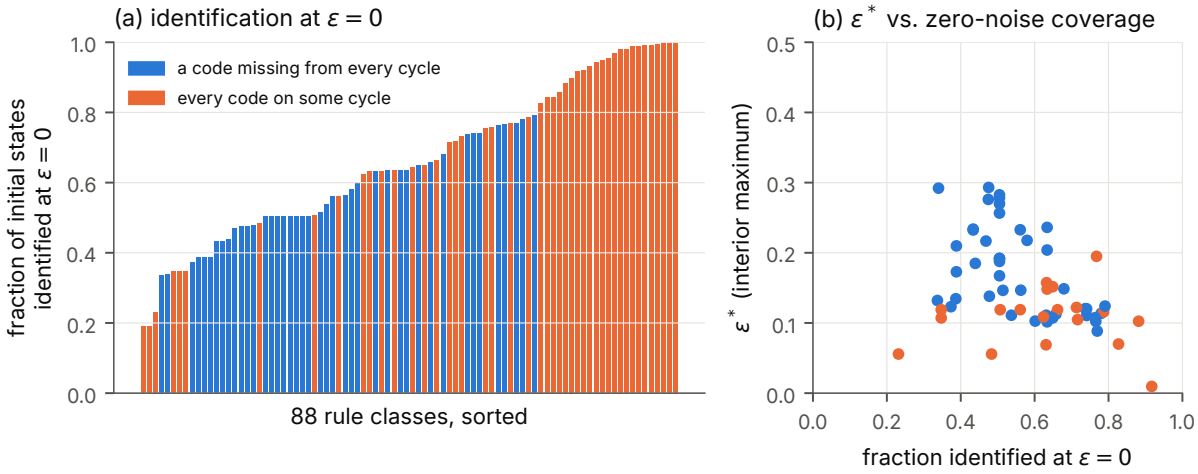


Figure 1: (a) Fraction of the 4096 initial states from which the orbit determines the rule ($|C_\infty| = 1$), for each of the 88 rule classes, sorted. Blue: some neighbourhood code occurs on no cycle of the deterministic map (42 classes); orange: every code occurs on some cycle (46 classes). (b) For the 61 classes whose rate curve falls towards $\varepsilon \to 0$ over the computed grid and has an interior maximum: the best noise level ε^* against the zero-noise identification fraction. Class 106 falls over the computed grid, but its rate diverges as $\varepsilon \to 0$ (Section 5); its point marks a local maximum.

class representatives, 13 of 1496 entries are unresolved (rules 7, 19, 23, 232, all at $\varepsilon \le 0.010$). The largest residual $\|\pi - \pi P\|_1$ among the dense solutions is 4.7e-13; the largest deviation between the dense solution and any power iteration among resolved entries is 5.2e-08.

Best noise level. For every class whose grid maximum of ρ is interior, ε^* is located by golden-section search between the neighbouring grid points to a width of $5 \cdot 10^{-4}$, each evaluation being a dense solve, and the solution at ε^* is verified as above.

Symmetry check. The chain is equivariant under reflection and complementation, so ν_c for the image of a rule under either symmetry is a permutation of ν_c for the rule itself (for a self-symmetric rule a constraint on its own ν_c). All 256 rules were computed independently; over the 12,972 pairs of a rule, a symmetry and a grid point at which the class representative is resolved, the largest deviation from the predicted permutation is 5.9e-11.

Zero noise. For each of the 256×4096 pairs (f, x_0) the orbit is followed until it closes; this yields $\tau(f, x_0)$, $N_\infty(f, x_0)$ and $|C_\infty|$ exactly.

Finite horizon. $\sum_{t<T} \mathbb{E}_f[n_c(X_t) \mid x_0]$ for all 4096 initial states at once is computed backwards, $G_t = P_{f,\varepsilon} G_{t-1}$ with G_0 the 4096×8 matrix of counts, using the tensor-product structure of the noise kernel; this is exact and costs $O(8n\,2^n)$ per step.

Analytic check. For rule 0 the chain is a sequence of independent Bernoulli(ε) vectors, so $\nu_7 = n\varepsilon^3$ exactly; the computed values deviate from $12\varepsilon^3$ by at most 4.1e-14.

The computation of version 0.1 runs in about two hours on two CPU cores; the additions of Section 5 took 1.4 h more in a from-scratch run on two cores. Code, data and a SHA-256 manifest of the 1011 data files accompany the note.

3 Results

3.1 Zero noise

Over all 1,048,576 pairs (f, x_0), 701,944 (66.9%) determine the rule; the others are stuck in a class of size $2, 4, \ldots$ forever ($|C_\infty| = 1$: 66.9%, $|C_\infty| = 2$: 21.6%, $|C_\infty| = 4$: 6.5%, $|C_\infty| = 8$: 3.4%, $|C_\infty| = 16$: 1.4%, $|C_\infty| = 32$: 0.1%, larger: 0.07%). The mean class size is 1.94, the largest 128. Orbits close quickly: the median of τ is 12 steps and the maximum over all pairs is 240 (class 45), so for $T \ge 240$ nothing in this section depends on T. The best classes identify the rule from 99.8%

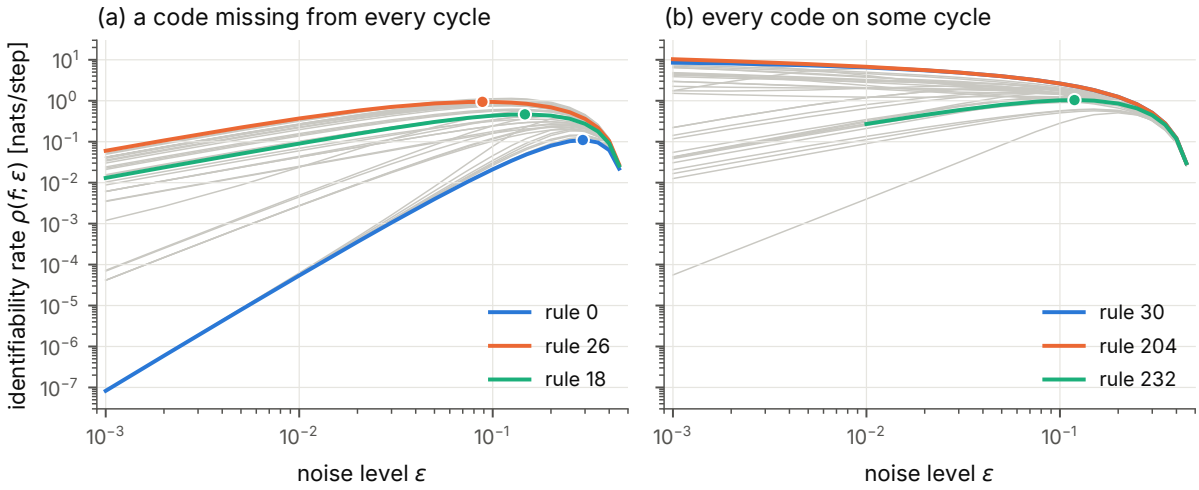


Figure 2: Identifiability rate $\rho(f;\varepsilon) = \lambda(\varepsilon)\min_c \nu_c(f;\varepsilon)$ for all 88 classes (grey) with three named rules per panel; dots mark ε^*. (a) Classes in which some code occurs on no cycle: $\rho \to 0$ at both ends (Lemma 3). (b) Classes in which every code occurs on some cycle: the curve either still has an interior maximum or keeps rising as $\varepsilon \to 0$; the eight coinciding top curves are the affine rules with uniform three-cell marginals, $\rho = \frac{n}{8}\lambda(\varepsilon)$.

of initial states (rules 25 and 60), the worst from 19.0% (rules 170, 204; these include the identity rule 204, whose every state is a fixed point and shows the observer exactly the neighbourhoods it starts with).

The structural quantity behind Lemma 3 is the set of codes that occur on cycles. In 42 classes (138 rules) some code is absent from every cycle; in the remaining 46 classes (118 rules) every code occurs on at least one cycle, but in none of them does every cycle cover all eight codes (0 classes do), so the small-noise limit of these classes depends on which cycles the noise selects.

3.2 Rates and the best noise level

Figure 2 shows the exact rate curves. All 42 classes with a code absent from every cycle have an interior maximum and fall towards $\varepsilon \to 0$ (the ratio of ρ at the two smallest resolved noise levels is at least 1.662), as Lemma 3 requires. Among the 46 classes whose cycles jointly cover all codes, 19 also fall towards $\varepsilon \to 0$ over the computed grid and have an interior maximum: for 18 of them the noise-selected cycles do miss a code, so they behave the same way; class 106 falls over the computed grid, but its rate diverges as $\varepsilon \to 0$ (Section 5). The other 27 rise as ε decreases (ratio in $[0.850, 0.986]$), consistent with the unbounded branch of Lemma 3. 8 of the rising classes (rules 15, 51, 60, 90, 105, 150, 170, 204) have $\nu_c = n/8$ for every code at every grid point (largest deviation 4.2e-12), hence $\rho(f;\varepsilon) = \frac{n}{8}\lambda(\varepsilon)$ (10.339 at $\varepsilon = 0.001$, where $\lambda = 6.893$); these eight curves coincide. All eight global maps are affine over GF(2). For 4 of them (rules 15, 51, 170, 204) the map is a bijection of S and the uniform law is stationary. For the other 4 (rules 60, 90, 105, 150, the additive rules, with kernels of dimension 1, 2, 2, 2 on the ring of 12) the stationary law is not uniform (its smallest entry at $\varepsilon = 0.1$ is 1.3e-04), but its three-cell marginals are: for $F_f x = Lx + b$ the stationary law satisfies $\hat{\pi}(\chi) = (1-2\varepsilon)^{|\chi|}(-1)^{\chi\cdot b}\hat{\pi}(L^\top\chi)$ in the Walsh basis, so $\hat{\pi}(\chi) = 0$ unless some power of $L^\top$ annihilates χ, and the three-cell marginal at $(i-1, i, i+1)$ is uniform iff $\hat{\pi}$ vanishes on the nonzero characters supported there; for the four rules the generalised kernel of $L^\top$ contains 0 such characters (checked by enumeration). On the grid every curve is monotone or unimodal except 2 (rules 41, 108), which rise towards $\varepsilon \to 0$, dip, and have a local maximum in between; unimodality is observed, not proved.

For the 61 classes that fall towards $\varepsilon \to 0$ over the computed grid, the interior maxima lie at $\varepsilon^* \in [0.010, 0.293]$ (median 0.132; smallest for rule 106, largest for rule 128; for class 106 the

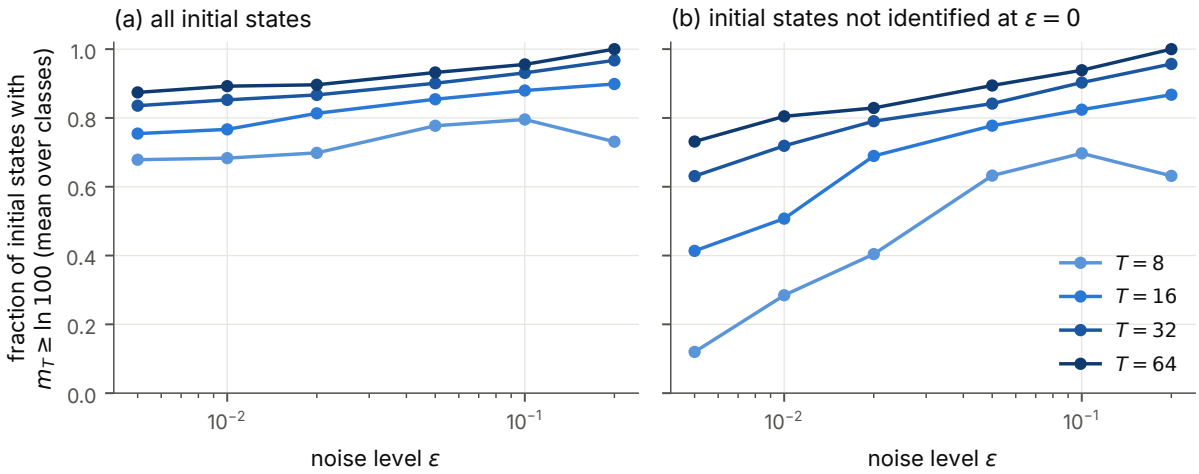


Figure 3: Fraction of initial states whose evidence margin $m_T(f; x_0, \varepsilon)$ against the hardest alternative reaches $\ln 100 \approx 4.61$ nats within T steps, averaged over the 88 classes. (a) all initial states; (b) only the initial states that never identify the rule at $\varepsilon = 0$. Exact values for $\varepsilon \in \{0.005, 0.01, 0.02, 0.05, 0.1, 0.2\}$.

maximum is local only, since its rate diverges as $\varepsilon \to 0$, Section 5), with peak rates $\rho^* \in [0.1101, 6.189]$ nats per step (smallest for rule 0, largest for rule 106). Rule 0, whose only cycle is the all-zero state, is the extreme case: the rarest code is 111, which requires three adjacent flips, $\nu_7 = 12\varepsilon^3$, and the maximum sits at $\varepsilon^* = 0.292$ with $\rho^* = 0.1101$ nats per step—at that noise level the observer needs about 9 steps to accumulate one nat of evidence against the rule that differs from 0 on 111. Rules 30 and 110 belong to the rising classes: their rates are 8.517 and 4.479 nats per step at $\varepsilon = 0.001$ and 2.637 and 1.484 at $\varepsilon = 0.1$; for them noise only lowers the rate. All classes are listed in Appendix A. Figure 1(b) relates ε^* to the zero-noise coverage over these 61 classes (class 106 included with its local maximum): the classes whose orbits identify the rule from fewer initial states tend to need more noise (Spearman rank correlation -0.44 with the identification fraction, $p = 4 \cdot 10^{-4}$; $+0.64$ with the mean class size, $p = 4 \cdot 10^{-8}$). At ε^* the rarest code is 111 for 41 of the 61 classes and 000 for 5: three adjacent equal cells are what noise has to create.

3.3 Finite horizon

The rate is an asymptotic statement; Figure 3 shows the exact finite-horizon margins from every initial state. All percentages in this paragraph are means over the 88 classes of the per-class fraction of initial states. At $\varepsilon = 0.005$ and $T = 8$, 67.9% of initial states have $m_T \geq \ln 100$; at $T = 64$ the fraction is 87.4%. Restricted to the initial states that are permanently non-identifiable at $\varepsilon = 0$—which have $m_T = 0$ for all T without noise, and of which a class has between 10 and 3316—the fraction rises from 12.0% ($T = 8$) to 73.1% ($T = 64$) at $\varepsilon = 0.005$, and at $\varepsilon = 0.2$ from 63.2% to 100.0%. Pooled over all such initial states instead of averaged over classes, the $T = 32$ fractions are 51.8% at $\varepsilon = 0.005$ and 93.2% at $\varepsilon = 0.2$ (class averages 63.1% and 95.7%). The noise level among the six computed that maximises the worst-case margin over initial states at $T = 32$ is, per class, $\varepsilon = 0.01$: 2, $\varepsilon = 0.02$: 12, $\varepsilon = 0.05$: 10, $\varepsilon = 0.1$: 32, $\varepsilon = 0.2$: 32 (median $\varepsilon = 0.1$, against a median $\varepsilon^* = 0.132$ for the interior maxima of the rate): at a horizon of a few dozen steps the transient, not the stationary frequency, decides how much noise is useful.

4 Boundary check: a generator outside the family

The definitions assume $f \in \mathcal{R}$. To see what the observer sees when this fails, three radius-2 rules were run through the same chain: the majority of five cells, the parity of five cells, and rule 110 with its output inverted on the single five-cell context 10101 (a rule that is radius-1 except on one context).

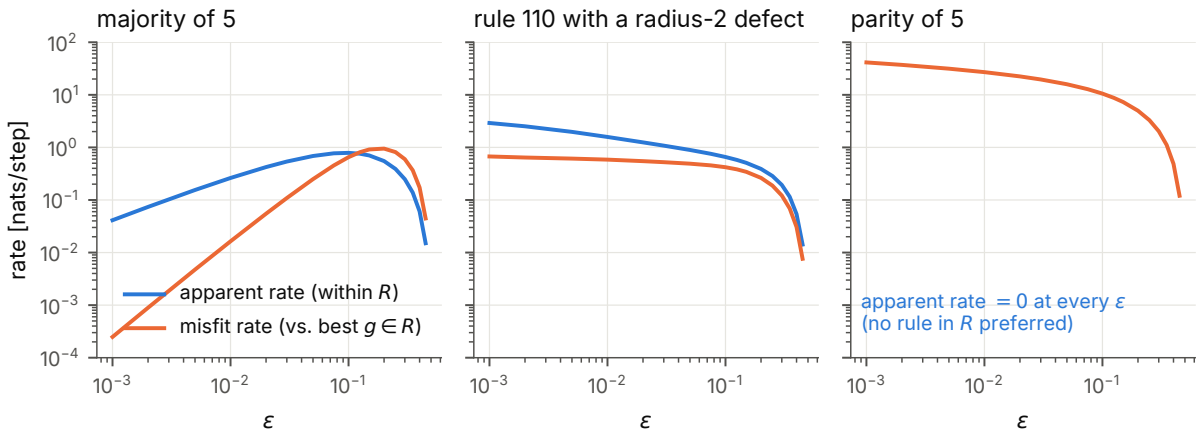


Figure 4: Three radius-2 generators scored by the radius-1 observer. Blue: the rate at which the posterior over $\mathcal{R}$ concentrates on the best rule $g^* \in \mathcal{R}$ (apparent identifiability). Orange: the relative-entropy rate between the true chain and the chain of g^* (the evidence that the family is wrong, available to a goodness-of-fit test).

None is reducible to radius 1. Lemma 1 holds for any two deterministic maps under the same noise, so with $a_{c,v} = \mathbb{E}_\pi[\#\{i : \text{code}_i = c,\ (F_h x)_i = v\}]$ the best in-family rule is $g^*(c) = \arg\max_v a_{c,v}$, the posterior over $\mathcal{R}$ concentrates on g^* at the rate $\lambda(\varepsilon) \min_c |a_{c,1} - a_{c,0}|$ ("apparent" rate), and the relative-entropy rate between the true chain and g^*'s chain is $\lambda(\varepsilon) \sum_c \min_v a_{c,v}$ ("misfit" rate). At $\varepsilon = 0$ the question is whether the orbit is consistent with *some* radius-1 rule.

generator	silent at $\varepsilon = 0$	g^*	apparent/misfit at $\varepsilon =$ 0.001	0.01	0.2	misfit > apparent from $\varepsilon =$
majority of 5	32.6%	232	166.0	16.0	0.58	0.125
rule 110, defect on 10101	7.1%	110	4.3	2.7	1.48	–
parity of 5	1.6%	tie	0.0	0.0	0.00	0.001

Table 1: Silent: fraction of the 4096 initial states whose zero-noise orbit is consistent with some radius-1 rule, so that the misspecification is never detected. g^*: the radius-1 rule the posterior concentrates on (the same at every ε for the first two generators; a tie for parity). Ratios: apparent rate divided by misfit rate.

For the majority rule the observer converges to rule 232 (majority of three) and, at $\varepsilon = 0.01$, accumulates evidence for it 16.0 times faster than evidence that the family is wrong; from 32.6% of initial states the zero-noise orbit never reveals the misspecification at all, while from the others it is revealed within the first two transitions. For the defective rule 110 the observer converges to rule 110 at every noise level, the misspecification being visible only through the context 10101. Parity of five is the opposite extreme: the deterministic map is a bijection, the stationary law is uniform, every radius-1 rule mismatches half of the positions, the posterior over $\mathcal{R}$ does not concentrate at all, and the misfit rate is maximal. Within the family, identifiability is a rate that noise can raise; outside it, the same noise raises the apparent rate and the misfit rate together, and which one wins depends on the generator. The observer's posterior alone cannot tell the two situations apart; only a check of the residual flip rate against the declared ε can.

5 Noise-selected cycles

Resistances and potentials. For $0 < \varepsilon < \frac{1}{2}$ the kernel is $P_{f,\varepsilon}(y \mid x) = \varepsilon^{r(x,y)}(1-\varepsilon)^{n-r(x,y)}$ with the *resistance* $r(x,y) = H(F_f x, y)$, and $r(x, F_f x) = 0$. Let $C_1, \ldots, C_m$ be the cycles of F_f, $R(u \to v)$ the minimum total resistance of a path from u to v, $V(C, C') = R(C \to C')$, and $\gamma(C)$ the minimum weight of a spanning in-tree on the cycles rooted at C with edge weights V, the

stochastic potential (Young, 1993). A cycle is *noise-selected* if $\gamma(C) = \min \gamma$. Bagnoli et al. (2024) studied all 88 classes at $L = 17$ under a "vanishing" noise of at most one flip per step: their chain between attractors, built from single-flip jumps into basins, has the correct leading-order law on cycles whenever the single-flip graph between cycles has one closed class, which is then exactly the set of noise-selected cycles (all resistances between cycles are at least 1). Rebuilt from their description in the independent verification of this version (`VERIFICATION_v02.md`), this holds for 87 of the 88 classes at $L = 17$, and for the remaining class 232 their answer is correct by complementation symmetry; so their $L = 17$ table already determines the noise-selected cycles, and with them the split $k^* = 0$ / $k^* \geq 1$, for all classes at $L = 17$. This version adds the exact orders k_c and k^*, the orders of all stationary probabilities, the asymptotics of ρ, certificates, the ring sizes $n = 4, \ldots, 20$ including those where the single-flip chain fails (class 164 at $n = 16$, where its limit also weights a cycle that is not noise-selected), and the correction for class 106. Write $\operatorname{ord} g$ for the exponent of the leading power of ε in $g(\varepsilon)$.

Theorem 1. *For every state x, $\operatorname{ord} \pi_{f,\varepsilon}(x) = \min_C[\gamma(C) + R(C \to x)] - \min_C \gamma(C)$. With D_c the set of states containing code c, $k_c = \min_C[\gamma(C) + R(C \to D_c)] - \min_C \gamma(C)$ and $k^* = \max_c k_c$: $\nu_c(f;\varepsilon) \sim a_c \varepsilon^{k_c}$ with $a_c > 0$, and $\rho(f;\varepsilon) \sim a\, \varepsilon^{k^*} \log \frac{1}{\varepsilon}$. In particular $\rho \to \infty$ iff $k^* = 0$, i.e. iff the noise-selected cycles jointly contain all eight codes.*

Proof. By the Markov chain tree theorem (Pitman and Tang, 2018, Thm. 1.1), $\pi_{f,\varepsilon}(x)$ is a sum over spanning in-trees rooted at x of products of transition probabilities, divided by the sum over all roots. Each product is a polynomial in ε with lowest term $\varepsilon^{r(T)}$ and coefficient 1, so nothing cancels and $\operatorname{ord} \pi(x) = W(x) - \min W$, $W(x)$ the minimum resistance of an in-tree rooted at x. For the reduction to cycles, censor the chain on $K' = \{\text{cycle states}\} \cup \{x\}$: the censored law is π restricted to K' and renormalised; since $\pi(K') \to 1$ by Lemma 3, the renormalising constant has order 0 and orders on K' are orders of π. The censored transition probabilities are convergent power series in ε with positive leading coefficient whose order is the minimum resistance of a path avoiding K' in between. Apply the tree theorem on K'. Edges inside a cycle cost 0, so cycles contract: a minimum in-tree on the states of K' leaves every cycle not containing the root by at least one edge, and these edges contain an in-tree of the contracted graph of no larger weight, while an in-tree of the contracted graph extends to the states along the cycle edges at no cost. Replacing edge weights by shortest-path weights does not change the minimum in-tree value, since the union of the shortest paths used by an in-tree contains an in-tree of no larger weight. If x lies on a cycle C_x, then K' consists of the cycle states and the formula reads $\gamma(C_x) - \min \gamma$, because $\gamma(C_x) \leq \gamma(C) + V(C, C_x) = \gamma(C) + R(C \to x)$ for every C. Otherwise, in a minimum in-tree on $\{C_1, \ldots, C_m, x\}$ rooted at x, redirect all but one of the cycles pointing to x to the cycle C_x that x flows to at cost 0 (keeping C_x itself if it points to x, otherwise the cycle on C_x's branch); this costs no more since $V(C', C_x) \leq R(C' \to x) + R(x \to C_x) = R(C' \to x)$, creates no cycle, and leaves an in-tree on the cycles plus one edge into x. The converse inequality is concatenation of a tree and a path. Since a move of resistance 0 from a cycle state stays on its cycle, $R(C \to x) = 0$ iff $x \in C$; hence $\operatorname{ord} \pi(x) = 0$ iff x lies on a noise-selected cycle, and $k_c = 0$ for every c iff the noise-selected cycles jointly contain all eight codes. Finally $\nu_c = \sum_x \pi(x) n_c(x)$ is a rational function of ε with positive leading coefficient and $\lambda(\varepsilon) \sim \log \frac{1}{\varepsilon}$. □

Computation. $R(C \to \cdot)$ for all states is a 0-1 breadth-first search on the state graph (deterministic step cost 0, single bit flip cost 1), one per rotation orbit of cycles. All potentials $\gamma(C)$ are obtained in one pass by a root-free Edmonds contraction (Edmonds, 1967): it returns a laminar family of sets with dual values $y_X \geq 0$ and one tight selected edge per set, and $\gamma(C) = \sum_X y_X - \sum_{X \ni C} y_X$ for every C. The family is a certificate: tight selected edges forming one cycle inside every contracted set give an in-tree of that weight for every root, and dual feasibility gives the matching lower bound. All quantities are integers. For bijective F_f the uniform law is stationary and $k^* = 0$; for affine non-bijective F_f the Walsh identity of Section 3 decides uniform three-cell marginals at every n.

Verification. Before any class was classified, high-precision stationary laws (subtraction-free GTH elimination at $\varepsilon = 10^{-3}, \ldots, 10^{-12}$) reproduced the predicted order of every state for all 256 rules and $n \leq 10$ (2048 pairs), and minimum in-trees on the full state graph reproduced the reduction to cycles exactly (2392 cases). At $n = 12$ all 256 rules were computed independently, the 42 classes of Lemma 3 have $k^* \geq 1$ and the bijective rules $k^* = 0$. An independent checker accepted all 88 certificates at $n = 12$ (4 of them, for the bijective classes, stubs asserting bijectivity) and 898 certificates for $n = 4, \ldots, 14$; for larger n there are no certificates. A separate verification session recomputed a sample from the definitions with its own code, checked all certificates with its own checker and read the proofs (`VERIFICATION_v02.md`).

Results at $n = 12$. The 42 classes with a code on no cycle have $k^* = 1$ (29 classes), 2 (5) or 3 (8). Of the 46 classes whose cycles jointly cover all codes, 28 have $k^* = 0$ and 18 have $k^* \geq 1$; 87 of the 88 classes agree with the trend classification of Section 3. The exception is class 106: 35 of its 37 cycles are noise-selected and some of them contain every code, so $k^* = 0$; high-precision solves give $\min_c \nu_c \to 0.0472$. Its maximum at $\varepsilon^* = 0.0095$ is local: ρ exceeds $\rho^* = 6.189$ again only for ε below about 10^{-57}. Rules 7, 19, 23 and 232, whose trend rested on $\varepsilon \geq 0.01$, have $k^* = 1$; their noise-selected cycles are the uniform states (a fixed pair or the two-cycle between them).

Ring size. For $n = 4, \ldots, 20$, 20 classes have $k^* = 0$ and 55 have $k^* \geq 1$ at every $n \geq 8$; the other 13 (classes 22, 26, 35, 37, 43, 45, 60, 73, 90, 94, 122, 142, 154) change side with n, several of them according to the divisibility of n. For the affine classes 60 and 90 the change is proved.

Proposition 1. *For the classes of rules* 60 *and* 90 *and every $n \geq 4$, $k^* \geq 1$ iff n is a power of two. At $n = 3$ both have $k^* = 1$.*

Proof. Identify S with $\mathrm{GF}(2)[t]/(t^n - 1)$, cell i with t^i. Rule 60 is multiplication by $L = 1 + t$ and rule 90 by $t^{-1}(1 + t)^2$ (up to the orientation of the ring, which does not matter), both with $b = 0$. Write $n = 2^a m$ with m odd. If $m = 1$, then $(1 + t)^n = 1 + t^n = 0$, so L is nilpotent, F_f^n maps every state to 0, the all-zero state is the only cycle, the codes other than 000 occur on no cycle, and $k^* \geq 1$ by Lemma 3. If $m > 1$, then $t^n - 1 = (t^m - 1)^{2^a}$ and $1 + t$ is a simple factor of $t^m - 1$, so the generalised kernel of $L^\top$ (the $(1 + t)$-primary part, $L^\top$ being multiplication by the reflected polynomial, which has the same $(1 + t)$-part) consists of the multiples of $\big((t^m - 1)/(1 + t)\big)^{2^a}$, a polynomial of degree $(m - 1)2^a$. A nonzero character supported on three adjacent cells is a shift of a polynomial of degree at most 2 and can be such a multiple only if $(m - 1)2^a \leq 2$, i.e. $n = 3$. Hence for $n \geq 4$ not a power of two the three-cell marginals are uniform at every ε (Section 3), $\nu_c = n/8$ and $k^* = 0$. At $n = 3$ the computed orders are $k^* = 1$ for both rules. □

6 What is established, and what is not

Established exactly for $n = 12$: the zero-noise classes and closure times for all 256×4096 pairs; the rate curves $\rho(f; \varepsilon)$ on the grid for all classes, with the resolved/unresolved status of every entry; ε^* and ρ^* for every class with an interior maximum; the finite-horizon margins for all initial states at the listed ε and T; the boundary check. New in version 0.2, established exactly with checked certificates: the noise-selected cycles, k_c and k^* for all 88 classes at $n = 12$ (which settles the classification left open by version 0.1, including rules 7, 19, 23, 232, and moves class 106 to the rising side) and for $n = 4, \ldots, 14$; established exactly by the same solver, without certificates, for $n = 15, \ldots, 20$ (a sample independently recomputed in `VERIFICATION_v02.md`). Proved for all n: Lemmas 1–3 and Theorem 1; $k^* = 0$ for bijective rules. Proved for all $n \geq 4$: the power-of-two criterion for classes 60 and 90 (Proposition 1). Observed, not proved: unimodality of the curves; every other dependence of the classification on n, which is computed for $n \leq 20$ only. Not claimed: the classification for $n > 20$ outside the affine and bijective classes; the leading coefficient a of ρ, which decides where the asymptotic regime begins (for class 106 far below any practical noise level);

observers who do not know ε or see only part of the ring; generators outside the family beyond the three checks of Section 4.

The practical reading is narrow: for a deterministic system whose attractor exercises only part of its rule, a known amount of dynamical noise is an instrument, not a nuisance, and the right amount can be computed from the rule's stationary neighbourhood frequencies. For a system whose attractor already exercises the whole rule, noise is only a nuisance. Which of the two one is facing can be read off the stationary neighbourhood frequencies; whether the family is the right one cannot be read off the posterior over it.

Statement on the use of AI

The definitions, the code, the computation, the figures and the text of this note were produced by Claude (Anthropic) in sessions directed by the author on 4 and 15 October 2026. A separate Claude session that had not seen the work recomputed samples of every quantity from the definitions, checked every number in the text against the data files and read the proofs; its findings (among them a false claim that four additive rules are bijections) were corrected in version 0.1, and its report accompanies the code. The author selected the question, set the scope, and is responsible for the content. Section 5 and the other changes of version 0.2 were produced in the same way in a session on 8 November 2026, with predictions sealed before the solver was written and verifiers run before any class was classified. A separate Claude session that had not seen the work then reviewed version 0.2 on 9 November 2026: it sealed a sample before reading the claims, recomputed it from the definitions with its own code, checked all certificates with its own checker, read the proofs and the related work, and checked the text against the data files. It found no disagreement in any computed value and a number of errors in texts, claims and scripts; its report (`VERIFICATION_v02.md`, with code and outputs in `verify_v02/`) accompanies the code, and its header states how each finding was handled in this version.

Code and data

The archive accompanying this note contains the code (`code/`, Python with NumPy), all data files (`data/`, with `MANIFEST.sha256`), and a script that regenerates every number and figure in this note from the data files, so that no number here is typed by hand. Repository: `https://github.com/frnkptrln/eca-noise-identifiability`; version 0.1 is archived as doi:10.5281/zenodo.23399101, all versions under doi:10.5281/zenodo.23399100.

References

F. Bagnoli, M. Baia and T. Matteuzzi. Effects of a vanishing noise on elementary cellular automata phase-space structure. In *Cellular Automata (ACRI 2024)*, LNCS, Springer, 2024. doi:10.1007/978-3-031-71552-5_5.

J. Bretagnolle and C. Huber. Estimation des densités: risque minimax. *Zeitschrift für Wahrscheinlichkeitstheorie und verwandte Gebiete*, 47:119–137, 1979.

T. M. Cover and J. A. Thomas. *Elements of Information Theory*. Wiley, 2nd edition, 2006.

J. Edmonds. Optimum branchings. *J. Res. Natl. Bur. Standards*, 71B:233–240, 1967.

M. I. Freidlin and A. D. Wentzell. *Random Perturbations of Dynamical Systems*. Springer, 3rd edition, 2012.

W. Li and N. Packard. The structure of the elementary cellular automata rule space. *Complex Systems*, 4:281–297, 1990.

J. R. Norris. *Markov Chains*. Cambridge University Press, 1997.

J. Pitman and W. Tang. Tree formulas, mean first passage times and Kemeny's constant of a Markov chain. *Bernoulli*, 24, 2018.

A. B. Tsybakov. *Introduction to Nonparametric Estimation*. Springer, 2009.

S. Wolfram. Statistical mechanics of cellular automata. *Reviews of Modern Physics*, 55:601–644, 1983.

A. Wuensche and M. Lesser. *The Global Dynamics of Cellular Automata*. Addison-Wesley, 1992.

H. P. Young. The evolution of conventions. *Econometrica*, 61:57–84, 1993.

A All 88 classes

Columns: class representative f (smallest rule number in the class); type (B: some code occurs on no cycle, so $\rho \to 0$ as $\varepsilon \to 0$ by Lemma 3; U: every code occurs on some cycle); id.%: fraction of the 4096 initial states that determine the rule at $\varepsilon = 0$; $\overline{|C_\infty|}$: mean zero-noise class size; $\tau_{\max}$: longest transient plus period; cycles: number of cycles of F_f; $|\bigcup N|$: number of codes occurring on cycles; ε^*, ρ^*: location and value of the interior maximum of ρ ($\to 0$: the largest resolved rate is at the smallest noise level and still rising; †: a local maximum, the curve rises again as $\varepsilon \to 0$).

f	type	id. %	$\overline{\lvert C_\infty \rvert}$	$\tau_{\max}$	cycles	$\lvert\bigcup N\rvert$	ε^*	ρ^*
0	B	34.0	2.97	2	1	1	0.292	0.110
1	B	63.1	1.79	3	162	7	0.111	0.766
2	B	43.4	2.69	13	11	4	0.234	0.200
3	B	65.8	1.77	25	39	7	0.113	0.869
4	B	38.8	2.86	2	322	5	0.210	0.225
5	U	73.1	1.79	3	435	8	$\to 0$	–
6	B	51.4	1.80	18	28	7	0.146	0.485
7	U	88.2	1.40	31	10	8	0.103	1.604
8	B	50.5	2.32	3	1	1	0.283	0.143
9	U	99.0	1.08	26	14	8	$\to 0$	–
10	B	50.5	2.54	13	31	6	0.167	0.358
11	B	78.0	1.48	16	32	7	0.114	0.968
12	B	50.5	2.45	2	322	5	0.192	0.308
13	B	74.1	1.58	11	30	6	0.111	0.937
14	B	53.7	1.97	16	36	7	0.111	0.776
15	U	84.3	1.54	12	352	8	$\to 0$	–
18	B	56.2	1.97	38	20	7	0.147	0.465
19	B	73.7	1.72	4	162	6	0.120	1.039
22	U	95.5	1.29	24	23	8	$\to 0$	–
23	U	66.2	1.85	7	164	8	0.119	1.040
24	B	58.0	2.15	14	11	4	0.218	0.277
25	U	99.8	1.06	34	21	8	$\to 0$	–
26	B	77.0	1.50	33	64	7	0.088	0.941
27	U	94.9	1.19	29	44	8	$\to 0$	–
28	B	60.2	1.63	12	75	7	0.103	0.799
29	U	85.6	1.36	3	751	8	$\to 0$	–
30	U	99.5	1.09	228	12	8	$\to 0$	–
32	B	50.5	2.52	7	2	3	0.278	0.149
33	U	78.7	1.40	7	163	8	0.116	0.916
34	B	50.5	2.45	13	31	5	0.192	0.308
35	U	63.1	1.71	29	42	8	0.069	1.638
36	B	67.9	2.01	3	98	4	0.149	0.443
37	U	94.3	1.23	23	58	8	$\to 0$	–
38	B	76.6	1.40	14	52	7	0.102	1.016
40	B	50.5	2.44	12	5	5	0.257	0.224
41	U	98.8	1.20	38	28	8	$0.035^\dagger$	$2.287^\dagger$
42	B	33.7	3.02	13	132	7	0.132	0.626
43	U	48.3	2.11	15	61	8	0.056	2.062
44	B	79.1	1.74	7	99	7	0.124	0.760
45	U	99.7	1.09	240	33	8	$\to 0$	–
46	B	64.9	1.76	14	11	6	0.107	1.123
50	B	63.4	1.79	8	163	7	0.102	0.804
51	U	84.3	1.54	2	2048	8	$\to 0$	–
54	U	98.0	1.10	25	33	8	$\to 0$	–
56	B	47.8	2.55	17	34	7	0.138	0.582
57	U	71.6	1.68	20	8	8	0.105	1.263
58	U	62.4	1.83	27	9	8	0.109	1.178
60	U	99.8	1.08	16	24	8	$\to 0$	–
62	U	96.9	1.16	29	92	8	$\to 0$	–
72	B	56.1	2.37	3	98	6	0.233	0.250
73	U	89.8	1.22	14	162	8	$\to 0$	–
74	U	91.9	1.83	21	29	8	$\to 0$	–
76	B	38.7	2.94	2	1499	7	0.135	0.596
77	U	56.1	2.07	6	325	8	0.119	1.040
78	B	74.1	1.60	12	30	5	0.120	1.145
90	U	93.2	1.40	6	70	8	$\to 0$	–
94	U	82.7	1.53	12	147	8	0.070	1.511
104	U	76.8	2.15	10	50	8	0.195	0.508
105	U	98.9	1.13	6	10	8	$\to 0$	–
106	U	91.7	1.90	168	37	8	$0.010^\dagger$	$6.189^\dagger$
108	U	75.6	1.84	4	674	8	$\to 0$	–
110	U	99.0	1.09	44	11	8	$\to 0$	–
122	U	75.7	1.59	41	21	8	$\to 0$	–
126	B	76.4	1.40	38	20	7	0.108	0.862
128	B	47.6	2.53	7	2	2	0.293	0.130
130	B	63.4	2.27	17	12	5	0.236	0.244
132	B	46.8	2.58	6	323	6	0.217	0.264
134	U	63.4	1.55	19	29	8	0.148	0.609
136	B	50.5	2.34	12	2	2	0.270	0.201
138	B	37.4	2.82	13	77	7	0.123	0.821
140	B	50.5	2.47	11	323	6	0.188	0.450
142	U	23.1	2.39	15	61	8	0.056	2.062
146	U	63.2	1.68	39	21	8	0.157	0.567
150	U	97.9	1.15	6	10	8	$\to 0$	–
152	B	63.4	2.11	23	12	5	0.204	0.429
154	U	64.5	1.72	24	308	8	$\to 0$	–
156	U	34.7	1.90	11	76	8	0.107	1.253
160	B	47.5	2.66	7	3	4	0.276	0.188
162	B	43.9	2.72	17	32	6	0.185	0.430
164	U	64.9	2.16	6	108	8	0.152	0.625
168	B	43.4	3.14	12	32	6	0.232	0.386
170	U	19.0	3.93	12	352	8	$\to 0$	–
172	U	71.3	2.48	13	109	8	0.122	1.163
178	U	34.7	2.28	7	164	8	0.119	1.040
184	U	34.7	3.31	16	61	8	$\to 0$	–
200	B	38.8	3.14	2	853	7	0.173	0.591
204	U	19.0	3.93	1	4096	8	$\to 0$	–
232	U	50.7	3.33	6	325	8	0.119	1.040